

TULSI RAJNIKANT PATEL

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WORK EXPERIENCE

Machine Learning Engineer | Alkermes Inc., Waltham, MA (Remote) July 2025 – Present

- Developing EEG-based treatment prediction statistical **machine learning models** (Elastic Net, Random Forest, LightGBM) on patient recordings, applying PCA-based dimensionality reduction to cut feature space by 70% while retaining 95% of variance.
- Engineered model evaluation infrastructure with cross-validation, SHAP-based explainability reports, and **Power BI** data visualization dashboards, collaborating with a team of 4 data scientists to ensure FDA-compliant outputs via **MLflow**.
- Spearheaded development of a **full-stack** clinical report automation application with **Snowflake AI Cortex**, containerized with **Docker** and deployed on Snowpark Container Services via **CI/CD pipeline**, cutting drafting time by 80%.
- Architected a **RAG pipeline** using Snowflake Vector Store and **AWS S3**, embedding 100+ clinical documents and integrating **LLMs** (GPT-5, Llama-3.1-70B, Mistral Large) to auto-generate 10+ structured report sections with 90% summary accuracy.
- Designed an LLM-powered **chat assistant** allowing medical writers to refine report content via natural language prompts, reducing revision cycles by 65% and receiving a 4.7/5 usability score across 3 departments.

Research & Development Intern | Sahajanand Technologies Pvt. Ltd., Surat, India Jan 2023 – June 2023

- Developed a **Computer Vision** pipeline for diamond inclusion detection using **OpenCV** and **Scikit-image**, processing 500+ images with contour and segmentation techniques, improving detection accuracy to 93% versus an 80% manual baseline.
- Applied **preprocessing** techniques to handle variable image quality, and designed **image-processing pipelines** to extract facet and table zones, reducing region segmentation errors and cutting per-stone inspection time from 8 to under 2 minutes.
- Integrated **AWS DynamoDB** as the metadata store for grading results, enabling structured querying across 1k+ graded diamonds and reducing data retrieval latency, and delivering interactive **Power BI** dashboards for stakeholder reporting.

ML Researcher | Nirma University, Ahmedabad, India March 2021 – May 2023

- Orchestrated a large-scale **NLP** data pipeline using SNScrape and Google Translate API to collect 600K+ multilingual tweets from 523 politicians across 12 languages for a university-funded digital humanities project.
- Fine-tuned **Large Language Models** (BERT, RoBERTa) and **deep learning models** (CNN, LSTM, GRU) for sentiment analysis and gendered language classification, achieving 86% accuracy, a 14-point gain over the baseline.
- Delivered **Power BI** dashboard visualizing gendered language patterns, using **SQL** to aggregate tweet data to generate insights across the dataset, contributing to research that won 2nd Prize at Digital Humanities in Oxford Week at Oxford University.

Software Engineer Intern | Avioryn, Surat, India June 2022 – Aug 2022

- Co-developed core backend modules for 'CleanZone', a laundry logistics platform, using **Node.js** and **MongoDB**.
- Implemented OTP-based authentication via the **2Factor API**, QR code generation for product handoff, and a password recovery module in **JavaScript**, securing transactions and streamlining re-engagement for 30+ partner accounts.

PROJECTS

AI Clinical Assistant

- Built an end-to-end **RAG pipeline** using **OpenAI** text-embedding-3-small, semantically indexing 60K+ clinical records and storing 1.2M+ embeddings in **MongoDB Atlas**, reducing query response time from 4s to 1.5s via HNSW indexing.
- Deployed Mistral-7B-Instruct via **vLLM** with **Docker**, served through **RESTful API** with a **React + Tailwind CSS** frontend, achieving 91.5% accuracy and keeping hallucination rate below 5% on a 500-question evaluation set.

Amazon ML Challenge – Product Length Prediction

- Performed exploratory data analysis and feature engineering on 2M+ product listings using **PySpark** and **Pandas**, building distributed preprocessing pipelines to extract text and image signals for warehouse optimization.
- Developed a **multimodal CNN-ANN** model with **BERT** embeddings, outperforming Random Forest, XGBoost, and SVM baselines by 12% and reducing product length estimation error via hyperparameter tuning across 5+ model configurations.

TECHNICAL SKILLS

Languages/ Web Frameworks: Python, C/C++, Java, SQL, JavaScript, TypeScript, HTML/CSS, React, Node.js, Django, Flask, Streamlit

ML/AI Frameworks: Pandas, TensorFlow, PyTorch, Hugging Face, LangChain, Scikit-learn, OpenCV, Scikit-image

Cloud & DevOps: AWS (S3, DynamoDB), Docker, Kubernetes, PySpark, Snowflake, Snowpark Container Services

Databases: MySQL, PostgreSQL, MongoDB, MongoDB Atlas, Snowflake Vector Store

EDUCATION

Arizona State University

Master of Science in Computer Science

Aug 2023 – May 2025

Tempe, AZ | GPA: 4.00/4.00

Nirma University

B.Tech in Computer Science & Engineering | Minor in Finance

Jul 2019 – May 2023

Ahmedabad, India